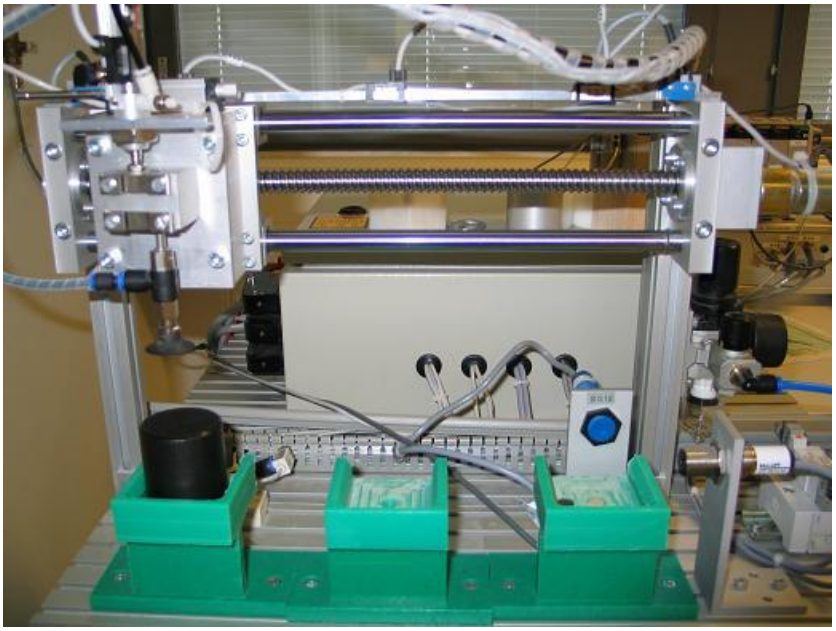
IDENTIFICATION STATION



1 Laboratory work in general

1.1 Preparations

Since all the laboratory works are available online, it is possible for the group to prepare the laboratory work beforehand. It makes the actual work in laboratory faster and easier if the group has done good preparations.

1.2 Pre-Works

1. The group explore this document contents and check out how the identification station in the laboratory works.
2. Open new project and adding I/O-list from the excel sheet.
3. The group makes manual drive part ready before proper laboratory work. Also touch screen manual control part making as ready as possible.

Additional the group should plan and program a roughly commented logic program, which accomplishes the functions told in the chapter 2 Description of activities.

The I/O list is also in the chapter 2.8, and you should use the I/O addresses listed there in your program. Plan also an I/O list for the communication between the touch screen and PLC. Use a proper memory area for that.

Use an example picture in chapter 2.8 to program the touch screen.

2 Function description

2.1 Logic control

Use ready FB:s (made in the Machine automation 1 course) as much as possible. Use at least two FC programs (ex. Main and Outputs). Make good comments in all networks. Put also all group members names in the top network.

2.2 Manual control

All the actuators of the identification station must be able to be controlled manually from the touch screen. Manual operations can be performed only when the Auto/Man switch is in manual position in the touch screen, MRA gives Manual Permit and the emergency stop switch hasn't been pressed.

The carriage can be controlled to left or right only when the gripper cylinder is in the upper position. The controlling of the carriage's motor works with two control relays by which the carriage can be moved in both directions. The carriage is moved by directional push buttons and the movement goes on as long as the push button is pressed down.
The gripper cylinder is controlled by a spring-return valve. If the valve receives a signal "1" the cylinder goes down and if the valve receives a signal "0" the cylinder goes up.
The controlling of the suction cup's suction operates by two separate control relays. A push button is made for the controlling of the suction. In the button two titles alternate; "Suction on" and "Suction off" depending on whether the suction is controlled on or off.
The actuators are controlled by push buttons made into the touch screen. A manual control is made for all the functions of the identification station. When a button is pressed the controlling goes on or off which means that the push button is connected via trigger fb or not depending on the situation. When the Auto/Man switch in the touch screen is in auto position it resets all the manual operations.
Notice! Only the emergency push button is operated from the push button box. Any other controlling is done from the touch screen.

2.3 Reference run

The general purpose of reference run is to run the controlled machine or system to a state where the PLC can detect it and from which the automatic run can begin in a controlled way. If the identification station is not in its reference place must the reference run be performed before starting the automation run.
When the Auto/Man push button is in auto position and the Start- button on the touch screen is pressed down will the PLC move the gripper to upper limit, shut down the suction cup suction and move the carriage to the delivery place 1 or 2 depending on

which place you have chosen to be the reference point. The delivery place 1 or 2 is searched by "Search FB". First driving the carriage to the right towards the delivery places. If the carriage can't find the delivery place 1 or 2 in 12 seconds will the carriage automatically be driven to the left when it will supposedly find the delivery place 1 or 2. This preceding situation takes place when the carriage is on the right side of the reference place in which case the carriage will be driven to the right-hand end line.
Now the reference control has been completed and the automatic control can be started. (=Ref_ok is "1" to MRA Fb input ➔Auto Permit)

2.4 Escort memory (This function are not in use at this moment)

2.4.1 General information about escort memory

The idea of the escort memory technique is to deliver all the data that is significant in production. This data can be for example how many holes are drilled in the workpiece, whereon the piece the holes are drilled, which color the product is painted, how the product is equipped etc.

When a workpiece arrives in a workplace the content of the memory button attached to it is being read. Based on the content of the memory button is decided which stages the workpiece needs to go through. When the stages are performed a new code needs to be written in the memory button so that the next workplace will know what is already been done to the workpiece and what is still left to do.

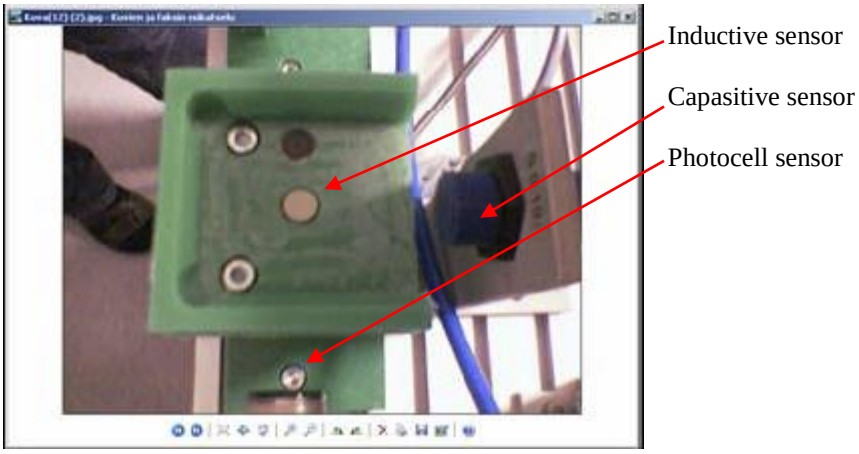
The absolute advantage of the escort memory technique is that the control system won't tangle even if pieces are added or removed in the production line. This is because the memory travels with the workpiece. Therefore the control system doesn't have to know and maintain the order of the work pieces on the production line.

The escort memory technique is used for writing or reading the identification code of the workpiece. The identification button is attached to the workpiece and it's read or written using the escort memory unit or an identification sensor.

2.4.1 Identification at this moment

The work piece identification works at this moment by tree normal sensors. There are tree sensors in the identification place. Photo-sensor give information that there is a piece. Inductive-sensor give information is the piece with or without metal. Capacitive-sensor give information material type or thickness. Picture 1. shows sensors places.

When the work piece on the id. place ➔ PLC program knows where this piece should be move.



Picture 1. Identification place with its sensors.

2.5 Shift register

At the moment when the work piece is being identified the sensors (inductive, capacitive, photocell). With Shift-register is possible to follow example what kind of work pieces are transferred. Example last 8 pieces. How this is realizable? This is a nut to crack for the group when all other things are ready ☺

2.6 Automatic run

When the Auto/Man switch is in Auto position, the Emergency-stop circuit hasn't gone off and the identification station is in its reference place can automation run be started by pressing the Start push button. Run the work cycle once. The automation run stops by pressing the Pause push button. When the Pause push button is pressed again will the identification station continue from where it was stopped.

<u>CYCLE</u>

- Detection station waits for the work piece to be identified (with or without metal plate)
- Press the Start push button when in Auto position
- Carriage from reference place to identification station
- Gripper down
- Suction on
- Wait until information for clipped on from the vacuum switch
- Gripper up
- Carriage in delivery place 1 or 2 (piece with or without metal plate)
- Gripper down
- Suction off
- Gripper up
- Back to the reference point

2.7 Alarms

The following alarms are made for the identification station:

Emergency-stop alarm

An alarm appears in the touch screen if the Emergency-stop button on the control box has been pressed.

Gripper cylinder malfunction

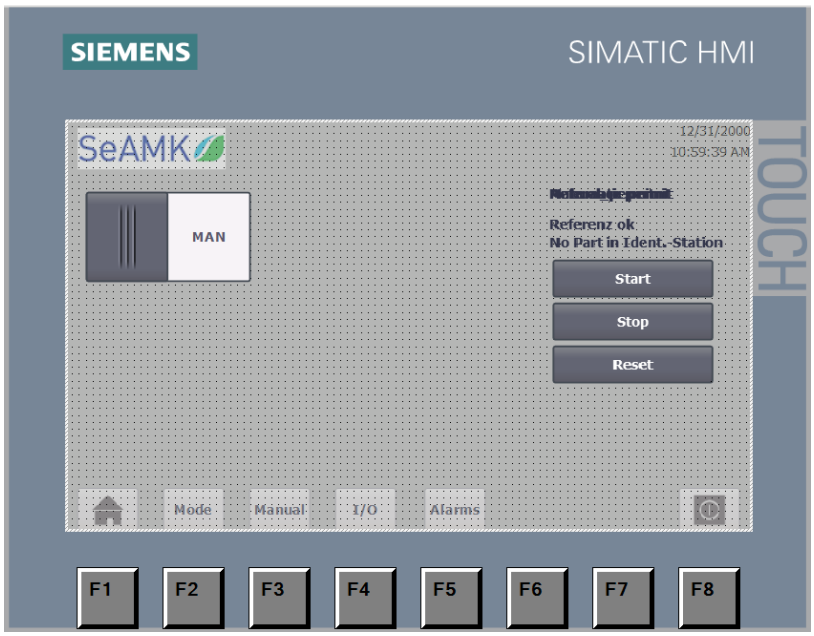
If the gripper cylinder doesn't reach its lower limit in 2 s, as the cylinder is being controlled downward, an alarm appears on the touch screen.
If the gripper cylinder doesn't reach its upper limit in 2 s, as the cylinder is being controlled upward, an alarm appears on the touch screen.

Escort memory alarm

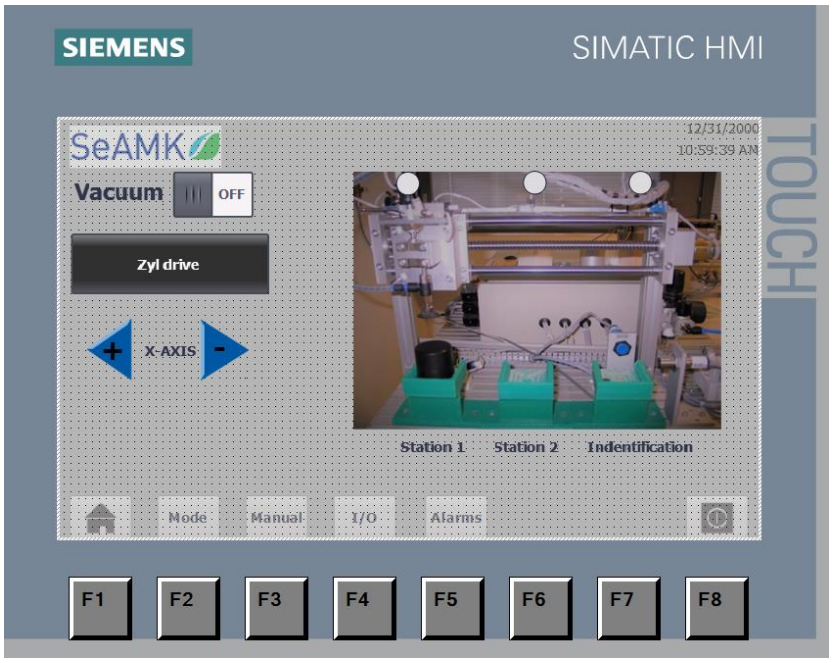
If the escort memory's ID-code is undefined an alarm appears on the touch screen.

2.8 Touch screen

Pictures 2 and 3 represent the touch screen's appearance and functions. The touch screen's appearance can also be personalized but the operations presented in the operation description must appear in it.



Picture 2. Operating select page (example)



Picture 3. Manual run page (example)

You can search help for example in the Ismo Tupamäki folder in lavitta. Koneautomaatio 1 MANUAALIT_Raportttia_varten /ohlog_osa_c.v19.pdf

2.8.5 Alarms

The Alarms -page is two-sheeted. This is done using the Frame function. Active alarms can be seen in the Alarms page and alarms history can be found in the Alarms history -page located beside. All active alarms can be signed for/deleted from the alarms page. Alarms and alarms history -functions are made using Alarm/Event Summary&History –display functions. Instructions for making the alarms –page can be found in the instruction guide ohlog_osa_c.v19.pdf pages 97-107.

page is used for writing the escort memory's ID-code and the other is used for reading the ID-code and for monitoring the states of the three other sensors at the moment of identification.

3 Laboratory work

Download the program you have done to the PLC. Read through the chapter 2 Description of activities, and make sure your program follows the instructions. If your program differs from the given instructions, do the required changes. Make sure that the teacher accepts your program.

Save your program carefully to the "sandbox"! \\cfile3\sandbox$\TupamäkiIsmo\

4 Specification

Because the specification is not required, **the program circuits must be clearly commented.** The laboratory work is accepted, when the teacher has checked the group's program in laboratory.

When your laboratory work is accepted in all its parts, your group will get a green mark in the duty list to the place of your laboratory work.

5 Documentation

5.1 Documentation in general

All the documentation is done with the same text layouts and instructions as the thesis work. Visit the link below to find the instructions for the thesis.
Behind the link you find the guidance for thesis in pdf-document, the empty thesis pattern with layout settings and also the cover page model.

http://intra.epedu.fi/Opiskelijaintra_%28SeAmk%29/Kansainvalisyys/Key_documents_for_international_students.iw3

5.2 Returning the specification

Required attachments for the specification:

- answers for the questions in chapter 4
- a working logic program in electronic form, which is commented and tested
- a working and tested program for the touch panel in electronic form

Compress all the returnable documents (specification, logic program, touch panel program) in one compressed file.

Return the file via students' Intra → "Harjoitustöiden palautus".

Return the specification within two weeks after completing the laboratory work.